

Euclid's Elements — Book II

Proposition 9

Formal Reconstruction — Technical Documentation

Jurek Rózański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJTEAMLAB

If a straight line is cut into equal and unequal segments, then the sum of the squares on the unequal segments of the whole is double the sum of the square on the half and the square on the straight line between the points of section.

1 Introduction

Proposition II.9 is a genuine architectural change in the current reconstruction of Book II. Propositions II.1–II.8 were organized around the reusable rectangle calculus and, in their configured public forms, concluded with exact `HilbertScissorsEq`. Proposition II.9 does not call any earlier Book II proposition. Instead, the current source deliberately returns to Euclid's Book I construction and angle argument and then passes through four applications of Proposition I.47 to the scissors/content layer.

Let AB be bisected at C , and let the unequal cut point D lie in the oriented branch

$$A - C - D - B.$$

The proposition asserts that the two squares on the unequal segments AD and DB have the same content as two copies of the square on the half AC plus two copies of the square on the segment CD between the two points of section. In modern notation the familiar shadow is

$$AD^2 + DB^2 = 2(AC^2 + CD^2).$$

This formula is only a mnemonic. The public Lean conclusion neither defines a numerical area nor multiplies segment lengths. It constructs four actual square representatives and relates their formal scissors sums by `HilbertScissorsEquicomplementable`.

The production source begins with the imports

```
import CGJteamLab.Proposition2_9Construction
import CGJteamLab.HilbertAngleDecomposition
import CGJteamLab.HilbertSquareTransport
```

```

import CGJteamLab.Proposition06
import CGJteamLab.Proposition32
import CGJteamLab.Proposition34
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47

```

The absence of a Book II import is intentional. The source comment describes the proof as the source-faithful synthetic proof of the oriented branch and records the classical route through I.5, I.29, I.32, I.34, I.6 and four uses of I.47, with explicit scissors bookkeeping.

2 Euclid's Statement and Classical Configuration

2.1 The statement

Euclid's Proposition II.9 says that if a straight line is cut into equal and unequal segments, then the sum of the squares on the two unequal segments of the whole is double the sum of the square on the half and the square on the straight line between the two points of section [?].

With the current lettering, C is the midpoint of AB and D is the second point of section. The formal theorem fixes the orientation $C - D - B$, so the unequal segments of the whole are AD and DB , the half is AC , and the segment between the points of section is CD . Thus the geometric statement is

$$\text{Sq}(AD) + \text{Sq}(DB) = 2 \text{Sq}(AC) + 2 \text{Sq}(CD),$$

where the equality is an equality of geometric content, not an equality of real-valued areas.

2.2 The classical construction

Euclid first erects CE perpendicular to AB at C and lays off

$$CE = AC = CB.$$

He then joins EA and EB [?]. This is the first complete auxiliary state: the midpoint and unequal cut are already fixed on the base, but no parallel through D has yet been introduced.

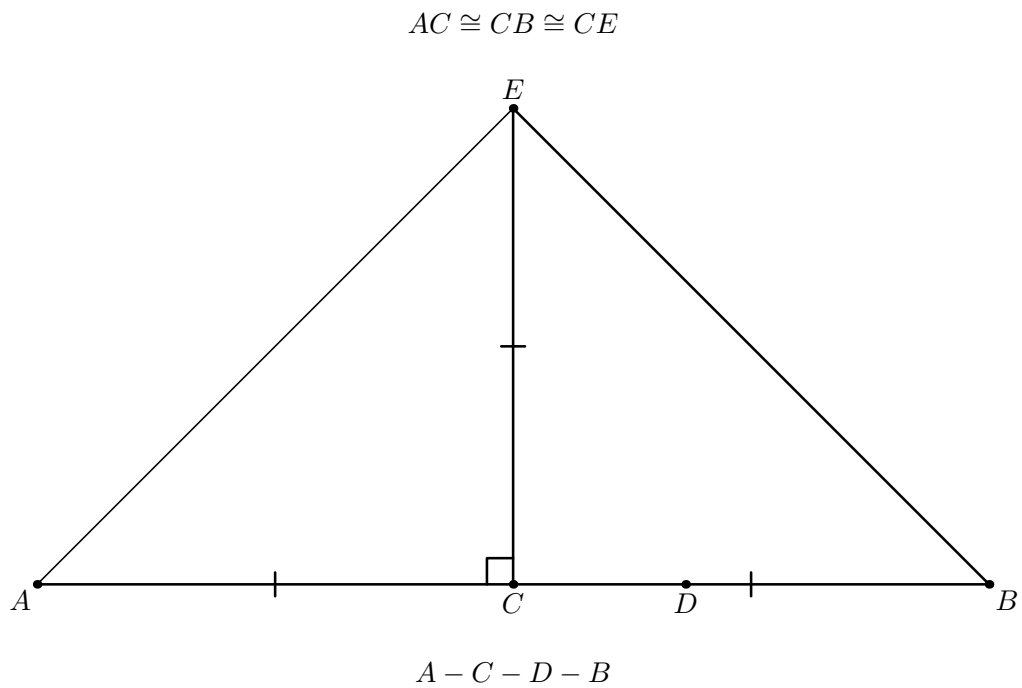


Figure 1: The first geometric state of II.9. The oriented base has order $A - C - D - B$, while CE is perpendicular to the base and satisfies $CE \cong AC \cong CB$. The joins EA and EB expose the two isosceles triangles ACE and CEB used in the first angle block. The points F and G and all later metric conclusions are deliberately absent.

Through D Euclid draws the line parallel to CE until it meets EB at F . The formal construction must prove that the intersection is on the correct open segment of the carrier, namely

$$E - F - B.$$

This directional fact is later used to transport the right angle AEB to the same ray EF .

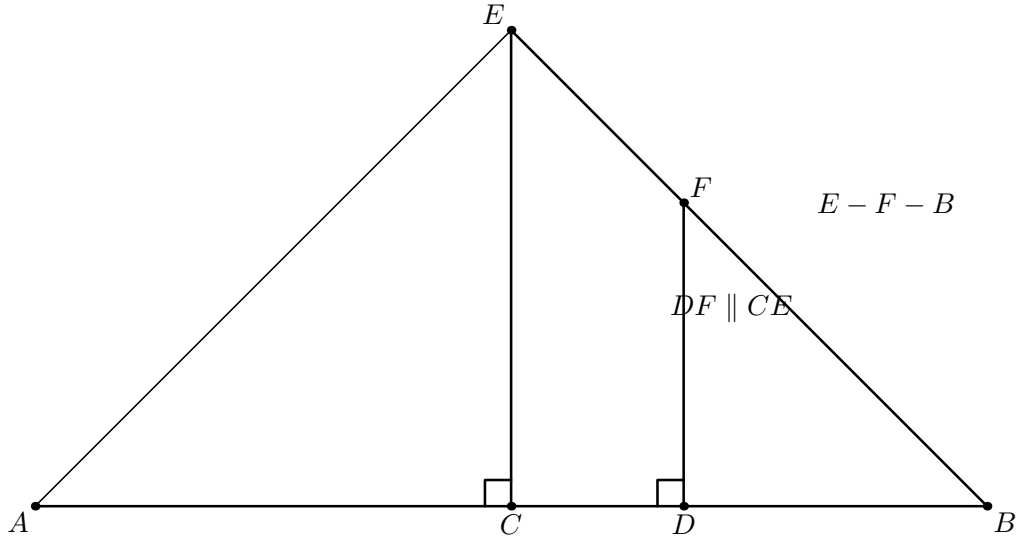


Figure 2: The second state of the II.9 construction. The line through D parallel to CE meets EB at F , and the strict order is $E - F - B$. The right angle at D is already forced by the parallel/perpendicular configuration. The point G has not yet been constructed, so no parallelogram or isosceles conclusion involving G is shown.

Through F draw FG parallel to the base. Its intersection with the carrier CE is the point G , and $DFGC$ is the auxiliary parallelogram. The current construction module does more than assert the intersection: a forced-Pasch argument proves

$$C - G - E.$$

Together with $E - F - B$, this is the complete directed source configuration on which the later angle transports depend.

Pasch fixes the order of G

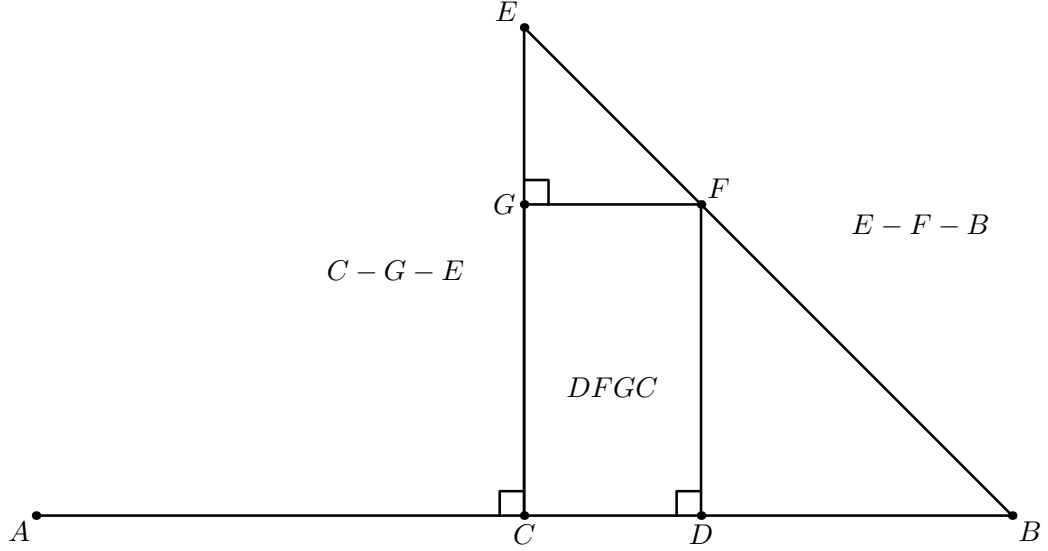


Figure 3: The third construction state. The horizontal through F closes the parallelogram $DFGC$ and meets CE at G . The figure records both strict orders $E - F - B$ and $C - G - E$. In the formal construction the second order is recovered by forced Pasch in the triangle EBC and line uniqueness. The later equalities $GE \cong GF$ and $DF \cong DB$ are not yet marked because they require separate angle arguments.

Finally Euclid joins AF [?].

Diagrammatic audit. The original single diagram displayed the finished construction together with metric conclusions that are proved only later. The audited sequence separates three genuine construction states: perpendicular-and-copy, the directed intersection $E - F - B$, and the Pasch-controlled completion $C - G - E$. The next figures are reserved for the angle consequences and the four Pythagorean blocks; representative transport in the scissors layer receives no geometric picture.

3 Classical Proof

The proof has two parts: first Euclid extracts the metric equalities required by the construction; then four Pythagorean applications identify one and the same square on AF in two ways [?].

Because $AC = CE$, Proposition I.5 gives equal base angles in triangle ACE . Since the angle at C is right, Proposition I.32 shows that each of those base angles is half a right angle. The same argument in triangle CEB shows that its two base angles are also half right angles. Hence the whole angle AEB is right, and because $E - F - B$, the angle AEF is right as well.

I.5 + I.32

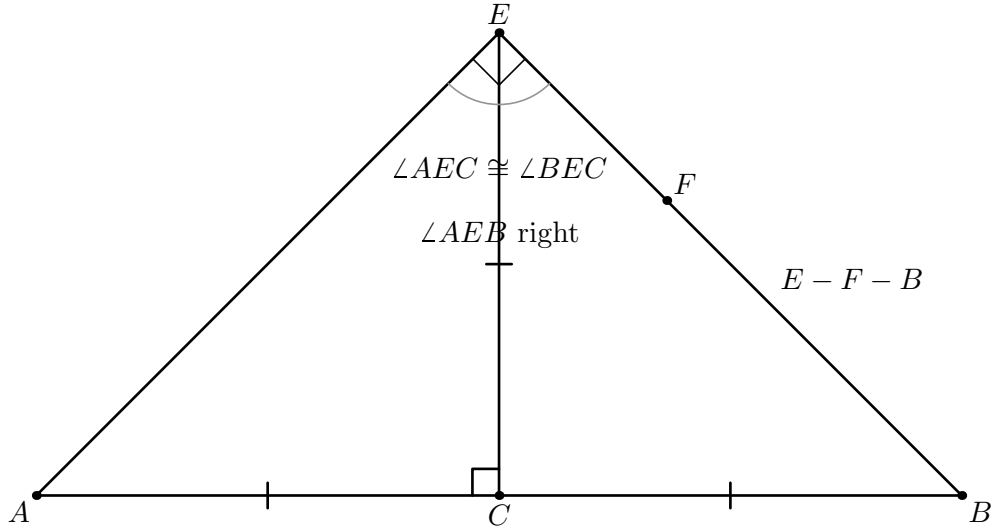
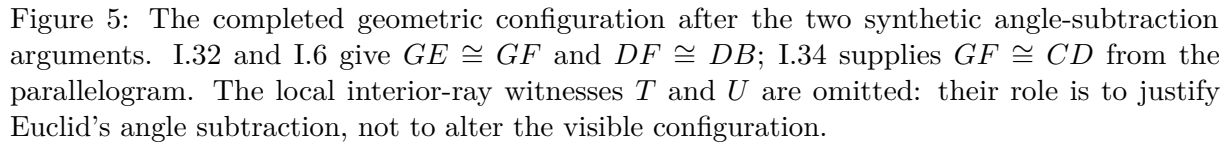


Figure 4: The first angle state of the proof. The isosceles triangles ACE and CEB give congruent angle components at E by I.5 and I.32. Their union is the right angle AEB , and the proven order $E - F - B$ transports that right angle to AEF . The formal witnesses R and S certify the angle decompositions but are omitted because they do not create new classical vertices.

$$GE = GF, \quad DF = DB.$$
$$GF = CD.$$

1.32 + angle subtraction + 1.6



Now apply I.47 four times. In triangle ACE , right at C ,

$$\text{Sq}(AE) = \text{Sq}(AC) + \text{Sq}(CE) = 2 \text{Sq}(AC).$$

In triangle EGF , right at G ,

$$\text{Sq}(EF) = \text{Sq}(EG) + \text{Sq}(GF) = 2 \text{Sq}(CD).$$

In triangle AEF , right at E ,

$$\text{Sq}(AF) = \text{Sq}(AE) + \text{Sq}(EF) = 2 \text{Sq}(AC) + 2 \text{Sq}(CD).$$

Finally, in triangle ADF , right at D ,

$$\text{Sq}(AF) = \text{Sq}(AD) + \text{Sq}(DF) = \text{Sq}(AD) + \text{Sq}(DB).$$

Comparing the two descriptions of the square on AF proves II.9.

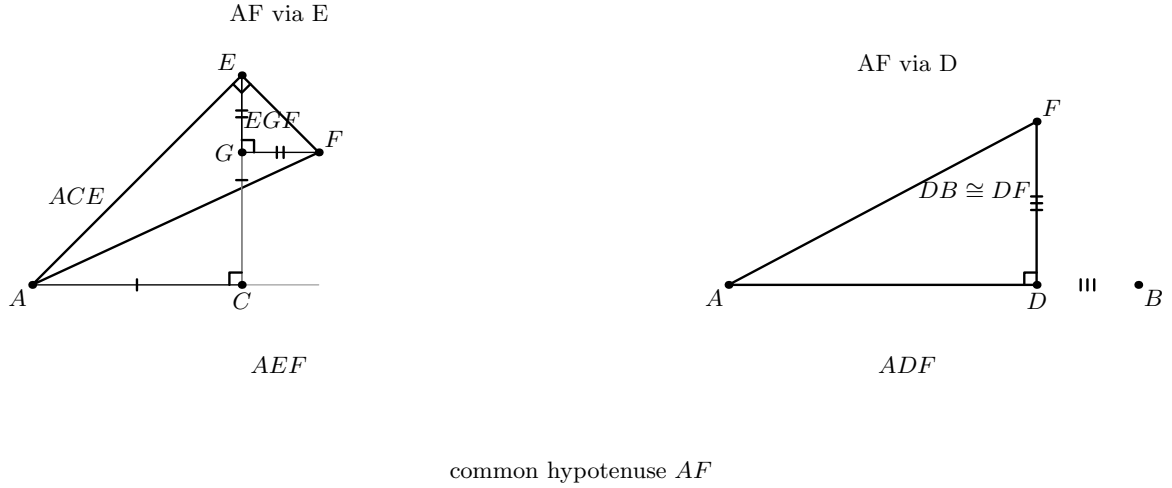


Figure 6: The four right triangles used by I.47, organized by the common square on AF . The left panel follows AF through the right triangle AEF , then expands its two leg squares with the right triangles ACE and EGF . The right panel gives the independent decomposition of the same hypotenuse through ADF . Segment congruences later transport DF to DB , the two $A/C/E$ legs to AC , and GE, GF to CD . No literal square representatives are drawn because the formal proof may construct them independently and then transport their content.

Diagrammatic audit. After Figure 6 there is no further geometric state to draw. Steps 5–7 of the formal architecture choose target square representatives, identify independent representatives of the square on AF , and compose equicomplementability relations. Those operations change the content bookkeeping, not the Euclidean configuration.

The classical local dependency picture is therefore

$$\begin{array}{c}
\text{I.11 + I.3: perpendicular and copied half} \\
\Downarrow \\
\text{I.31: parallel through } D \quad + \quad \text{join } EA, EB, AF \\
\Downarrow \\
\text{I.5 + I.29 + I.32 + I.6: } GE = GF, DF = DB \\
\Downarrow \\
\text{I.34: } GF = CD \\
\Downarrow \\
\text{four applications of I.47} \\
\Downarrow \\
\text{II.9.}
\end{array}$$

4 The Geometric Identity in the Current Formalization

The public theorem assumes

```

(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hCDB : Geo.Between C D B)

```

so the midpoint package supplies $A - C - B$ and $AC \cong CB$, while the second hypothesis selects the oriented branch $C - D - B$. The proof derives the required four-point order $A - C - D - B$ rather than taking it as a primitive datum.

The theorem then constructs concrete squares on AD , DB , AC , and CD . If their formal terms are denoted by

$$S_{AD}, \quad S_{DB}, \quad S_{AC}, \quad S_{CD},$$

the conclusion has the shape

$$S_{AD} + S_{DB} \simeq_{\text{ec}} (S_{AC} + S_{AC}) + (S_{CD} + S_{CD}).$$

Here $\simeq_{\text{ec}}$ denotes the relation formalized by `HilbertScissorsEquicomplementable`.

This distinction is essential. Unlike the configured theorems II.1–II.8, II.9 does *not* conclude `HilbertScissorsEq`. Nor does it conclude an equality of numerical areas. Its four Pythagorean blocks inherit the content strength of the current formalization of I.47 and are composed by symmetry, transitivity, additive compatibility, and transport between square representatives.

5 Reusable Infrastructure Used

5.1 The permanent II.9 construction layer

The file `Proposition2_9Construction.lean` is a separate geometric layer. Its public helpers progressively establish the source diagram rather than mixing construction with the final content

proof. In particular, `proposition2_9_G_between_CE` returns points E, F, G with

$$E - F - B, \quad C - G - E, \quad A - C - D,$$

together with the right angle at C , $CE \cong CB$, the parallelogram $DFGC$, and its opposite-side congruences.

The construction module itself imports I.31, I.34, and I.47 infrastructure. Its initial perpendicular/copy helper packages the I.11 + I.3 construction, while its later order proofs use the established Hilbert incidence/order and Pasch machinery to locate F and G on the correct open segments.

5.2 Angle decomposition rather than numerical angle measure

The source proof repeatedly says that an angle is half a right angle and later subtracts equal parts from equal right angles. The formal proof does not introduce degrees or real-valued angle measures. It uses `HilbertRayMeetsSegment` as the interior-ray witness and the reusable lemmas

```
hilbert_angleDecomposition_halves_congruent_of_whole_congruent
hilbert_angleDecomposition_angle_addition_interior
hilbert_angleDecomposition_angle_subtraction
hilbert_angleDecomposition_angle_subtraction_right
```

from `HilbertAngleDecomposition.lean`.

5.3 Square transport

The four I.47 calls do not automatically choose the same square representative on a common side. The proof therefore uses `hilbert_square_transport` to move between squares erected on congruent segments and between different representatives of the same segment. This is most visible for the two occurrences of the square on AF and the two occurrences of the square on EF .

This is representation transport, not a hidden equation $x^2 = y^2$ in a numerical ring. The transported objects remain concrete `hilbertParallelogramTerm` values in the scissors calculus.

5.4 Book I square construction and Pythagoras

The public theorem constructs fresh target squares on AD , DB , AC , and CD with `euclid_proposition_46`. It then imports four already proved I.47 decompositions and transports their square terms into those target representatives. Thus I.46 controls representation while I.47 controls the content identities.

6 Proof Architecture of the Reconstruction

6.1 Step 1: recover the right-angle configuration

`proposition2_9_right_angle_configuration` starts from the permanent construction and proves the right angles needed later:

$$\angle ADF, \quad \angle EGF, \quad \angle ACE, \quad \angle BCE.$$

The proof propagates right angles through same-ray transport, opposite extensions, and adjacent angles of the parallelogram $DFGC$.

6.2 Step 2: reconstruct Euclid's half-right-angle block

`proposition2_9_classical_angle_block` applies I.5 in the two isosceles triangles ACE and CEB , then applies I.32 in oriented exterior-angle form. The resulting points R and S lie on EB and EA and act as explicit interior-ray witnesses for the two angle bisections at C .

Using equality of right angles and the angle-decomposition API, the proof shows that the two components AEC and BEC are congruent. Their sum is therefore a right angle, giving AEB right; the same-ray fact $E - F - B$ then gives AEF right.

6.3 Step 3: recover the two isosceles triangles

`proposition2_9_classical_isosceles_blocks` proves

$$GE \cong GF, \quad DF \cong DB.$$

This is deliberately not done by segment subtraction. For each equality the proof follows the Euclidean angle route: create an I.32 decomposition witness, compare two right angles, subtract congruent components synthetically, obtain congruent base angles, and finish with `euclid_proposition_6`.

The first block introduces a witness T inside the right angle FGC ; the second introduces a witness U inside the right angle CDF . These points are formal witnesses for Euclid's angle arithmetic and are not needed in the final statement.

6.4 Step 4: build the four Pythagorean blocks

`proposition2_9_four_pythagoras_blocks` adds the I.34 equality $GF \cong CD$ and invokes I.47 on exactly the four right triangles used in the classical proof:

$$ACE, \quad EGF, \quad AEF, \quad ADF.$$

The helper returns all four existential square packages together with the segment congruences needed to identify their legs with the target sides.

6.5 Step 5: construct target squares

The public theorem uses I.46 four times to choose concrete squares $S_{AD}, S_{DB}, S_{AC}, S_{CD}$. This separates the statement-level representatives from whatever representatives happened to be produced inside the four I.47 applications.

6.6 Step 6: follow the common square on AF

The first content path begins with I.47 on ADF :

$$S_{AF} \simeq_{\text{ec}} S_{DA} + S_{DF}.$$

Square transport replaces DA by AD and DF by DB , yielding

$$S_{AF} \simeq_{\text{ec}} S_{AD} + S_{DB}.$$

The proof reverses this relation, moves to the AF representative used by the AEF block, and applies I.47 again to obtain

$$S_{AD} + S_{DB} \simeq_{\text{ec}} S_{EA} + S_{EF}.$$

6.7 Step 7: replace AE and EF by four leg squares

Representative transport first identifies the EA square with the AE square used by triangle ACE and the EF square with the EF square used by triangle EGF . The two remaining I.47 blocks then give

$$S_{EA} + S_{EF} \simeq_{\text{ec}} (S_{CA} + S_{CE}) + (S_{GE} + S_{GF}).$$

Finally the segment congruences

$$CA \cong AC, \quad CE \cong AC, \quad GE \cong CD, \quad GF \cong CD$$

transport these four squares to two copies of S_{AC} and two copies of S_{CD} . Additivity and transitivity close the public theorem.

7 Lean Formalization

7.1 The public theorem signature

The following is an ASCII transcription of the current public signature:

```
theorem euclid_proposition_2_9_oriented
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hMidC : HilbertIsMidpoint Geo C A B)
  (hCDB : Geo.Between C D B) :
  exists SADO SAD1 SDB0 SDB1 SAC0 SAC1 SCDO SCD1 : Geo.Point,
    IsSquare Geo A D SADO SAD1 /\
    IsSquare Geo D B SDB0 SDB1 /\
    IsSquare Geo A C SAC0 SAC1 /\
    IsSquare Geo C D SCDO SCD1 /\
    HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A D SADO SAD1 +
     hilbertParallelogramTerm Geo D B SDB0 SDB1)
    ((hilbertParallelogramTerm Geo A C SAC0 SAC1 +
     hilbertParallelogramTerm Geo A C SAC0 SAC1) +
     (hilbertParallelogramTerm Geo C D SCDO SCD1 +
     hilbertParallelogramTerm Geo C D SCDO SCD1)) := by
```

The suffix `_oriented` is substantive documentation. The public theorem proved in this file is the branch in which the unequal point D lies strictly between the midpoint C and endpoint B . The document must therefore not silently replace the formal statement by an un-oriented theorem over all possible positions of D .

7.2 The proposition-local theorem chain

The main source file is organized around four named proposition-local helpers:

```
proposition2_9_right_angle_configuration
proposition2_9_classical_angle_block
proposition2_9_classical_isosceles_blocks
proposition2_9_four_pythagoras_blocks
```

They form a genuine mathematical factorization of the proof. The public statement does not expose the auxiliary points E, F, G, R, S, T, U because they are construction and angle-decomposition witnesses rather than part of the final content identity.

7.3 The final content bridge

The last stage contains five conceptually different transports:

- (a) I.47 on ADF converts the left-hand target sum into a square on AF ;
- (b) two AF representatives are identified;
- (c) I.47 on AEF converts that square into squares on EA and EF ;
- (d) I.47 on ACE and EGF expands those two squares into four leg squares;
- (e) square transport identifies the four legs with two copies of AC and two copies of CD .

No cancellation of content terms occurs. The proof is a forward chain of synthetic equicompleteness relations.

8 Relationship with Earlier Book II Propositions

There is no earlier Book II theorem in the production dependency chain for II.9. In particular, the theorem does not call II.1, II.4, II.7, II.8, or the `HilbertRectangle` rectangle-contained-by-calculus that dominated the previous block.

This is not merely an omission at theorem-call level. The production file imports `Proposition2_9Construction`, the angle-decomposition and square transport layers, and Book I propositions I.6, I.32, I.34, I.46, and I.47. The construction file itself imports Book I I.31, I.34, and I.47 infrastructure. No `Proposition2_N` module with $N < 9$ appears in these imports.

Mathematically, II.9 therefore begins a new Book II proof mode in this project: its algebraic-looking square identity is reconstructed from the classical Euclidean configuration instead of being normalized through the previously built rectangle identities.

9 Relationship with Book I Infrastructure

The classical and formal DAGs are close in mathematical shape but not identical line-by-line.

Classically, the construction uses I.11 and I.3 for the perpendicular and copied half, I.31 for the parallel, I.5 for isosceles base angles, I.29 for parallel angle transport, I.32 for the triangle angle-sum/exterior-angle mechanism, I.6 for the converse isosceles conclusions, I.34 for opposite sides of the parallelogram, and I.47 four times [?].

The Lean development packages some of these into reusable lower-level infrastructure. For example, “half a right angle” becomes an explicit interior-ray decomposition, and Euclid’s subtraction of equal angle parts is realized by angle-decomposition lemmas. Conversely, Lean adds an explicit use of I.46 to construct the four square representatives that occur in the public statement.

Thus the source-faithful mathematical DAG is retained, but the proof is not a literal transcription of Euclid’s prose.

10 Formalization Notes and Hidden Geometric Data

10.1 The oriented branch is part of the theorem

The hypothesis `hCDB : Geo.Between C D B` records the choice of the right-hand unequal cut. Euclid’s printed statement is symmetric in the two possible unequal cuts, but the current public theorem proves this one branch. Any later symmetric wrapper should be documented as an additional theorem, not read into the present signature.

10.2 Betweenness controls same rays

The order statements $E - F - B$, $C - G - E$, $A - C - D$, and their reversed forms are used to build `HilbertSameRay` witnesses. Those witnesses justify the replacement of an angle arm by a collinear point on the same ray. Lean cannot infer these transports from the appearance of the diagram.

10.3 The points R , S , T , U encode angle arithmetic

The exterior-angle theorem I.32 produces explicit points used as interior rays. The first pair R, S produces the two half-right-angle decompositions needed to prove AEB right. Later T and U support the two synthetic angle-subtraction arguments that end in $GE = GF$ and $DF = DB$.

This is the main hidden geometric data of II.9. It replaces numerical statements such as “ $45^\circ + 45^\circ = 90^\circ$ ” and “subtract equal angles from equal right angles” by incidence, betweenness, ray, and angle-congruence witnesses.

10.4 Noncollinearity is pervasive

Every right-angle transport, angle congruence, and application of I.6 or I.47 requires nondegeneracy. The proof repeatedly derives noncollinearity from the parallelogram, the perpendicular

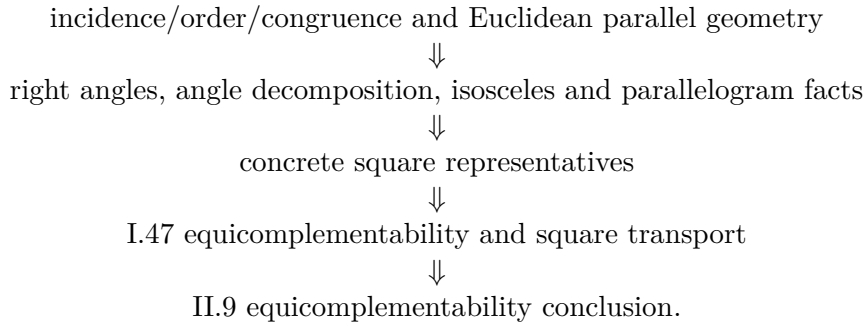
construction, strict betweenness, and collinearity transitivity. These obligations are mathematically meaningful but are largely suppressed by the classical diagram.

10.5 Square identity is weaker than exact scissors equality

The final relation is equicomplementability. It is therefore incorrect to describe II.9 as an exact cut of the left pair of target squares into the four right-hand target squares. The proof composes the equicomplementability data returned by I.47 with square-representative transport. The appropriate prose is “equal content/equicomplementable”, not “the same dissection”, unless a later theorem strengthens the result.

11 Axiomatic and Semantic Status

The current source introduces no numerical area function and no multiplication of segment lengths. Its semantic layers are



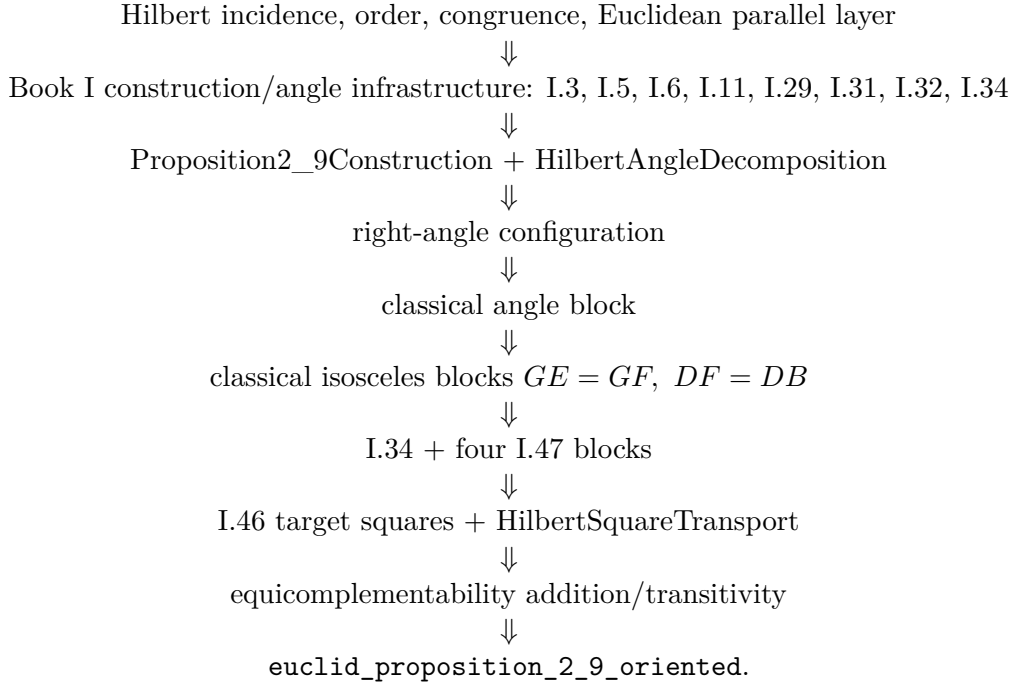
A source inspection of `Proposition2_9.lean` shows no local `axiom`, `sorry`, or `admit` declaration. The present documentation pass does not contain a recorded output of

```
#print axioms Geometry.euclid_proposition_2_9_oriented
```

so no transitive axiom list is asserted here. This distinction is deliberate: “no new local axiom” is weaker than “axiom-free”, and a final audit should record the actual `#print axioms` output when it is run in the project environment.

12 Dependency Summary

The production chain is



The implementation checkpoint from the inspected source is:

- new proposition-specific geometric axiom declaration: no;
- new proposition-specific content axiom declaration: no;
- proposition-local helper theorem declarations: yes;
- separate reusable infrastructure used: HilbertAngleDecomposition and HilbertSquareTransport;
- new HilbertRectangle theorem used or required: no;
- earlier Book II proposition used: no;
- conclusion is exact HilbertScissorsEq: no;
- conclusion is HilbertScissorsEquicomplementable: yes;
- uses numerical area: no;
- uses segment multiplication: no;
- local sorry/admit in Proposition2_9.lean: no;
- transitive #print axioms audit: not recorded in the materials available to this documentation pass.

13 Conclusion

Proposition II.9 is the first clear break in the current Book II documentation from the rectangle-normal-form strategy of II.1–II.8. Its formal proof is built around Euclid’s own auxiliary perpendicular, parallel, and right-triangle configuration. The angle argument is reconstructed synthetically, with interior-ray witnesses replacing numerical angle arithmetic, and the two key isosceles equalities are recovered through I.32, angle subtraction, and I.6.

The four applications of I.47 then expose the real content architecture. One square on AF is simultaneously connected with the two target squares on AD, DB and with the four leg squares coming from ACE and EGF . Congruence and square transport identify those legs with two copies of AC and two copies of CD .

Accordingly, the modern identity

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

should remain exactly what it is: a useful algebraic shadow. The theorem proved by Lean is a synthetic statement about order, perpendiculars, parallels, congruent segments, square representatives, and `HilbertScissorsEquicomplementable`; it neither invokes the earlier Book II rectangle calculus nor crosses into numerical area or segment multiplication.